

Project Summary: Bilingual AI Search Infrastructure for B2B Marketplace

Core Business Activity: What the Company Does tablazat.hu is a specialized B2B (business-to-business) matchmaking platform in Hungary that acts as an intelligent digital assistant connecting companies with active purchasing intent to verified, reliable suppliers. The platform's primary mission is to accelerate, streamline, and bring transparency to corporate procurement and sales processes.

Instead of being just a static directory, the platform operates a sophisticated ecosystem:

- **For Buyers:** It provides a **completely free** suite of procurement tools. This includes an AI-powered Request for Quotation (RFQ) Assistant, semantic search across 35,000+ categories, and verified financial and legal data (via Dun & Bradstreet) to support risk-free, value-based decision-making.
- **For Suppliers (Sellers):** It delivers pre-qualified, high-intent business leads (RFQs), thereby reducing customer acquisition costs and increasing sales efficiency.

The Business and Sales Model (Monetization) The company operates a two-sided marketplace based on a "**Freemium**" model.

1. **The Magnet (Demand Generation):** The buyer side is 100% free. By offering free AI tools, market research, and expert concierge support, the platform attracts a massive user base (over 12,000 registered buying companies, including large enterprises). This free side generates the high-quality demand and concrete RFQs that drive the business.
2. **The Engine (Revenue Generation):** The platform generates revenue exclusively from the supplier side through a premium, subscription-based (SaaS) service called the **Premium Partnership (VIP)**.

The Basis of Sales:

The Premium Partnership The core of the company's sales strategy is converting free or new suppliers into Premium Partners. The key characteristics of this monetization model include:

- **Fixed Subscription Fee, No Commissions:** It is an annual, subscription-based service. Pricing is predictable and based on the number of product categories (topics) the supplier chooses to dominate. The platform specifically avoids commission-based systems to maintain transparency and predictability.
- **Exclusive Value Proposition for Sales:** Suppliers buy the Premium Partnership because it guarantees them first-page visibility in search results (limited to a maximum of 8 spots per topic) and **exclusive, priority access** to all relevant, pre-qualified RFQs in their chosen categories.
- **Proactive Support & Retention:** To ensure high renewal rates, Premium Partners receive a "done-for-you" concierge service (where tablazat.hu builds and manages their digital storefronts) and a dedicated strategic account manager to help them maximize their ROI.

In short, tablazat.hu creates value by making procurement effortless and free for buyers, and monetizes this generated demand by offering guaranteed visibility and priority lead access to suppliers through a fixed-fee annual subscription

The platform manages a massive directory of over 38,000 B2B categories and 5,000 to 10,000 company profiles. Their business model revolves around capturing user intent, processing Requests for Quotations (RFQs), matching these RFQs to the most capable suppliers (using both internal database records and scraped web content), and plan to provide whitelabel chatbot solutions for partner websites.

The Business Needs and Problems

The company faces several critical challenges in their search and matching processes:

1. The "Problem vs. Solution" Semantic Gap: Buyers often search by describing a problem (e.g., "I need to renovate my warehouse lighting"), while suppliers list solutions and technical products (e.g., "LED fixtures"). Traditional search struggles to connect the two.
2. Linguistic Complexity (Hungarian Agglutination): The primary language is Hungarian, which is highly agglutinative (e.g., the word "targonca" can become "targoncákát"). Standard search engines fail to match these variations properly.
3. The Flaw of Pure AI (Vector) Search: While AI vector embeddings are great for understanding intent, they fail at exact keyword matches. For example, a search for a "5-ton forklift" might return 3-ton or 7-ton forklifts, and a search for a specific brand like "Bunting" or a certification like "ISO 9001" might return generic safety gear or quality management companies.
4. Data Quality & Duplication: The existing 38,000+ categories contain duplicates, spelling errors, and fragmented text (HTML tags, missing context) that confuse search algorithms.
5. Data Integrity: Updating large company profiles in the database can cause system crashes or partial data exposure if a process fails mid-run, leading to corrupt search results.
6. We also need to make the category search available in a Dify chatbot as there will be the proposal submitted and all the necessary information collected by the bot. The bot can't raise a good question about the product / service specification before knowing the correct category.

Expected results

The expected result of this project is a fully deployed, production-ready, bilingual (Hungarian and English) AI search infrastructure designed for a B2B marketplace.

Ultimately, this system will transform the platform's static database into an intelligent B2B matching engine capable of bridging the "semantic gap" (understanding complex buyer intent) while maintaining strict precision for exact technical keywords and brand names.

The project will deliver two independent, Dockerized software components, along with strict operational and performance guarantees:

1. The ETL Pipeline (Data Preparation & Indexing) A robust Python-based pipeline that will ingest 38,000+ category records and 5,000–10,000 company profiles exclusively from your internal PostgreSQL databases. The expected outputs of this pipeline include:

- **AI Enrichment & Narrative Templates:** The pipeline will clean HTML, translate missing language content, and use GPT-4o-mini to generate "Narrative Templates" (natural language sentences) and synthetic buyer questions to enrich the data context.
- **Strict Dual-Path Indexing:** It will prepare the data for a pgvector-enabled schema using two parallel paths. The *raw, natural text* will be sent to OpenAI to create dense vector embeddings (for semantic understanding), while the text will be separately lemmatized (using **huspacy** for Hungarian and **spaCy** for English) to generate BM25 tokens (for exact keyword matching).
- **Data Integrity & Idempotency:** The pipeline will feature hash-based change detection so it only incurs API costs for records that have actually changed. It will also use transactional "Atomic Swaps" (DELETE + INSERT) to ensure the database is never left in an empty or corrupt state if the pipeline crashes mid-run.

2. The Search Microservice (FastAPI) A low-latency REST API that will execute the hybrid search logic. Its expected capabilities include:

- **Reciprocal Rank Fusion (RRF):** It will execute the vector similarity search and the BM25 lexical search in parallel, mathematically blending the two ranked lists together using an RRF scoring formula (with a configurable *k* constant).
- **Three Core Endpoints:**
 - **/search/category:** A high-speed endpoint (under 500ms) for the central marketplace chatbot to instantly identify user intent. Connected to Dify chatbot.
 - **/match/companies:** A deep-matching backend engine that processes Requests for Quotations (RFQs), applies filters (like logistics capabilities), incorporates a "Partner Boost" multiplier, and returns the best-suited suppliers.
 - **/search/single-company:** A strictly isolated whitelabel search endpoint that powers chatbots embedded directly on partner websites.

3. Quality and Performance Targets By the end of the engagement, the system is expected to pass rigorous acceptance criteria, including:

- **The "Bunting Test":** A mandatory "Protected Terms Pass" ensuring that whitelisted brand names (e.g., "Bunting") and technical codes (e.g., "ISO-9001") are masked and perfectly preserved during lemmatization, ensuring accurate keyword searches.
- **High Accuracy:** Achieving a minimum of 88% accuracy for general category/company searches and 90% accuracy for specific brand/product searches.
- **Bilingual Parity:** The system must automatically detect query languages, handle Hungarian agglutination properly, seamlessly trigger cross-language fallbacks, and return equally high-quality results whether a user searches for "Targonca" or "Forklift".

Category	Metric / Requirement	Target / Threshold
Speed	Category Search Response Time (p95)	< 500ms
	Company Matching Response Time (p99)	< 5000ms
	Health Check Response Time	< 50ms
Connection	Database Connection Pool	Min 5, Max 20 connections
	Health Check Verification	Must verify DB, Vector Index, BM25, and API connectivity
Accuracy	ETL Processing Success Rate (Categories & Companies)	≥ 85%
	Category & Company Search Accuracy (General)	≥ 88%
	Brand & Product Search Accuracy	≥ 90%
	Technical Specification Search Accuracy	≥ 85%
	Language Detection Accuracy	> 95%
	Translation Quality (Technical term preservation)	> 95%
	Protected Terms Preservation & AI Enrichment Fields	100%

Acceptance Criteria: